

1 Subsequent Assignment of Status Reasons on AIP2

Overview

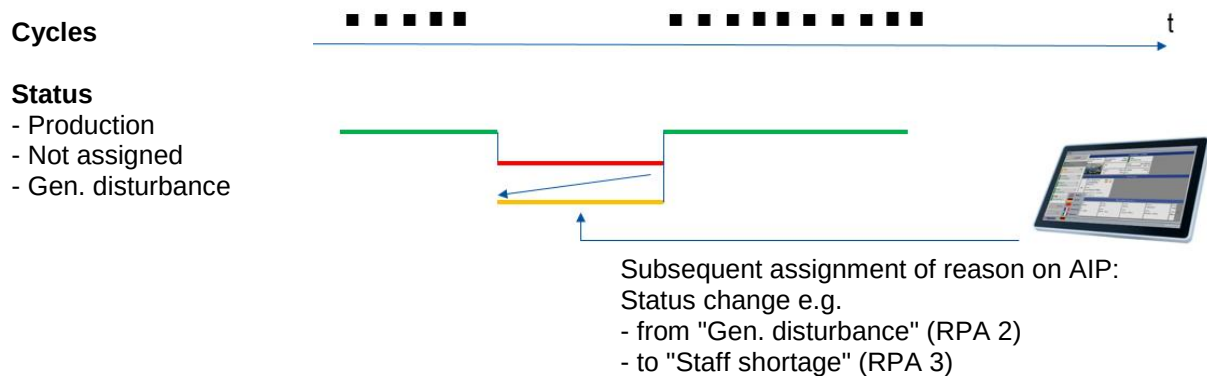
With a monitoring of machines, the following situation is possible:

Because of a malfunction, no cycles or no operating signal is sent to the PCC. The PCC identifies that signals are missing and automatically changes to status 30000 "Not assigned".

When the malfunction is finished, the machine control transfers signals to the PCC again. The PCC identifies the signals and automatically changes to status "Production". For the time of the malfunction, the system automatically posts the status "General disturbance". The status "General disturbance" is usually assigned to the resource performance account 2 (DCI).

The machine has restarted production and the operator then goes to the AIP to subsequently set the correct status. This status can be a status that is also assigned to the resource performance account 2. But it can also be a status that is assigned to another resource performance account.

The diagram below illustrates the subsequent assignment of a status reason, i. e. the status change from "General disturbance" (RPA2) to "Staff shortage" (RPA 3):



Purpose

You use this functionality in the following situation:

- You use the automatic status monitoring:
 - Cyclic monitoring
 - Monitoring via operating signal
- You use the function *Status log* on the AIP to assign a reason to the status "General disturbance" while the operation is still logged on.

- You want that the system updates the BDE and WRM result records when the status "General disturbance" is subsequently changed and the new status is assigned to a different resource performance account.

If this situation exists, the system automatically recalculates the postings of this workflow. After recalculation, the resource performance accounts in the BDE and WRM result records then correspond to the resource performance accounts of the workplace/machine statuses.

Integration

Note: This automatic recalculation and the subsequent assignment of a status reason is performed while the affected objects – e.g. operation, person, resource – are still logged on. The posting workflow is still active then and has not been closed.



The operator can subsequently change the status "General disturbance" in the *Status log* on the AIP. Condition: the current posting workflow must still be active.

Requirements

To use this functionality, the following requirements must be fulfilled:

- The functionality is available as of MW 4.0_{pe}.
- In the INI configuration, the SYSTEM entry "EVENT_MAINTENANCE" with key "EXTENDED" and value "J" must be active.

Note: The entry is created via the database patch INTEGRATED_DATA_MAINTENANCE (included in MW 4.0_{pe}).

- The database table cyclic_recalculation must exist.
 - The table is created via the database patch INTEGRATED_DATA_MAINTENANCE (included in MW 4.0_{pe}).
- In the scheduler, the entry "IDM cyclic recalculation" must be available and active.
 - In the scheduler, the entry "IDM cyclic recalculation" is made three times by the database patch. These entries trigger the automatic recalculation right after each shift change.


```
hymw.exe -u9999 -c"DLG=IDM.CYCLRECALC|RWSC.TIMEOUT=3600|"
```
 - The recalculation is performed at fixed times: at 06:05, 14:05 and 22:05 hours.



All three entries are not active by default. They must be activated **explicitly**. Then restart HYDRA.

Processing

Subsequent Assignment of Status Reasons on AIP

If the operator changes the status "General disturbance" using the AIP function *Status log* and confirms the input dialog, the system processes as follows:

- The AIP sends the status change to the HYDRA server. Command:
DLG=M_MST|MNR=<workplace/machine>|MST=<new status>|MOD=K].
- The parameter MOD=K informs the system that this is a subsequent assignment of reason and that it is not the change of the current status.
- The system identifies the MDE result record and replaces the previous status (and its resource performance account) in the MDE result record with the new status (and its resource performance account).
- If the new and the old status are assigned to the same resource performance account, the processing is completed.
- If the new status is assigned to a different resource performance account than the old status, the system identifies the event (M_MST) that was used to generate the MDE result record. The system writes the ID of the event identified into the internal table *cyclic_recalculation*.

In both cases, for the operator the processing on the terminal is finished: the new status is shown in the status log.

Recalculation of the saved status changes

The scheduler requests a web service. To ensure that the posting workflows are completed, the scheduler includes three entries and each entry requests the web service a few minutes after shift change. By default, the entries are at 06:05, 14:05 and 22:05 hours.



Change the entries in the scheduler, if the shifts in your company do not start at 06:00, 14:00 and 22:00 hours.

The web service selects all events saved in the table *cyclic_recalculation* and passes the selected events to the system for automatic recalculation. Beforehand, the system checks if or that the posting workflow is actually closed (see below).

The automatic recalculation process is identical to the recalculation function called in the tabular *Event maintenance*.

After processing of the automatic recalculation, the entries are deleted in the table *cyclic_recalculation*.

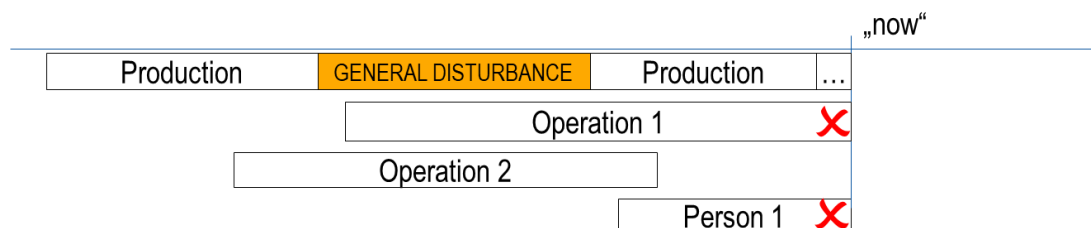
Checking whether the posting workflow has been completed

Each event identified has been performed at a specific workplace. A posting workflow is only completed for the event identified, if one of the following conditions is fulfilled:

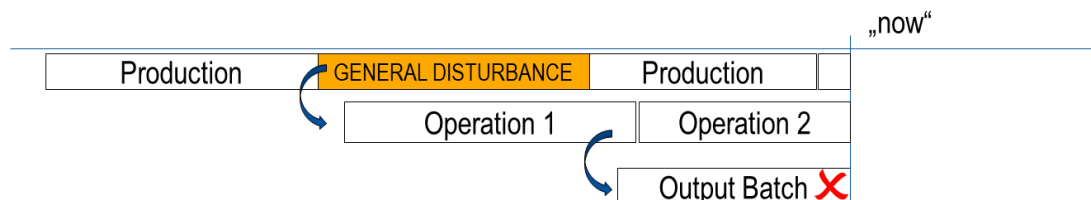
- The beginning of the current shift is after the time of the event.
- The beginning of the current shift is before the time of the event and the following two conditions are true:
 - a) At the workplace, there is no current (still running) logon available, which starts before the end of the status with a subsequently assigned reason.
 - b) At the workplace, a current (still running) logon is not available that has started in the period of time of a BDE log record generated in the meantime (record type U/E/B/H) and which is (entirely or partly) included in the period of time of the status with a subsequently assigned reason.

In the examples of posting workflows below, the relevant posting workflow is not yet completed and closed because of the data record with the red X.

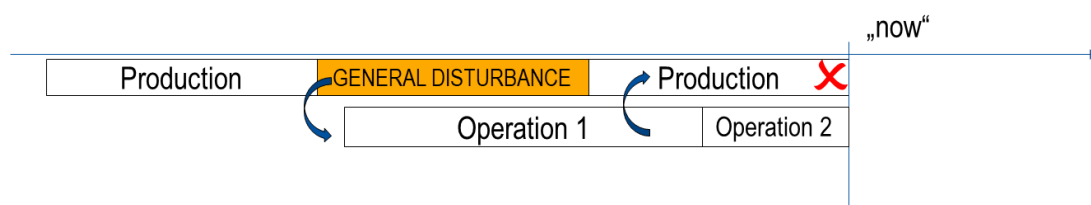
Example 1:



Example 2:



Example 3:



Further notes

In the course of the subsequent calculation and correction of values, also the changed MDE log record is canceled and a new MDE log record is created.